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How Effective are Public Policies to Increase Health Insurance Coverage Among Young Adults?†

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This paper assesses the impact of policies to increase insurance coverage for young adults. The introduction of SCHIP in 1997 enabled low-income teens up to age 19 to gain access to public health insurance. More recent policies enabled young adults between the ages of 19 and (typically) 24 to remain covered under their parents' health insurance. We use the discrete break in coverage at age 19 to evaluate the impact of SCHIP, and quasi-experimental variation to evaluate the impact of "extended parental coverage" laws. Our results suggest that both types of policies were effective at increasing health insurance coverage. (*JEL* G22, H75, I18, J13)

For the past 25 years, public policy regarding health insurance in the United States has focused on increasing the rate of coverage among children and youth. These policies appear to have been successful as rates of insurance coverage have increased substantially for these groups, as we discuss in greater detail below. The next frontier in insurance coverage is young adults.[1]

One option is to expand existing public insurance programs that are targeted at children and teens, such as Medicaid and the State Children's Health Insurance Program (SCHIP), by increasing the maximum age to qualify for coverage. Children generally lose their Medicaid or SCHIP coverage on their nineteenth birthday, but recent proposals have sought to increase this age limit to as high as 25 years old.[2] A second option is to expand private insurance coverage among young adults by increasing the age of dependency for the purposes of health insurance coverage. Due to incentives of the tax code, children generally lose access to private insurance coverage through their parents at age 19 (or age 24 if they are

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[1] This paper was completed prior to he passage of national health care reform, but the lessons learned here are relevant to the implications of the reform for coverage outcomes.

[2] See, for example, the Children's Health First Act (H.R.1535 and S.895), introduced in March 2007 by Rep. John Dingell and Sen. Hillary Clinton.

full-time students).[3] However, in recent years, several states have required insurance policies to allow young adults through (typically) age 24 to be covered under their parents' employer-provided plan, in spite of the tax implications.

This paper provides evidence on the efficacy of these two policy options to increase health insurance coverage among young adults. While the existing literature has examined the take-up and crowd-out rates associated with SCHIP, it is unclear that these estimates are applicable to young adults. We provide new estimates of take-up and crowd-out, which are more relevant to the young adult population, by taking advantage of the discontinuity in SCHIP eligibility upon turning age 19 that was generated when the program was instituted in the late 1990s. "Extended parental coverage" laws are recent enough that no past research of which we are aware has examined their effect on take-up and crowd-out. We take advantage of variation across states and years in the introduction of these laws, along with other provisions that target certain population subgroups, to identify the effects of these extended parental coverage laws.

Our estimates suggest that both policies have been successful in increasing insurance coverage for older teens and young adults. We find that SCHIP increased insurance coverage by 3 percentage points for older teens as a whole. Those under 150 and those between 150 and 300 percent of the poverty line experienced 7 and 4 percentage point increases, respectively. These estimates reflect roughly a 20 percent reduction in rates of uninsurance among these two groups. We find weaker evidence of crowd-out than in the existing SCHIP literature.

We also find that extended parental coverage laws are effective at increasing rates of insurance coverage. Laws implemented at the state level would lead to a 3 percentage point reduction in rates of uninsurance among those eligible, whereas a federal law would lead to a 7 percentage point reduction. The differential is attributable to a feature of ERISA, which exempts firms that self-insure from state insurance regulation. We find that the impact of extended parental coverage laws is at least partially attributable to the increased propensity of young adults to live with their parents when these laws are passed. Our results also provide some suggestive evidence of reverse crowd out; increases in private insurance coverage may be at least partially offset by reductions in public insurance coverage.

I. Background

A. *Descriptive Statistics Regarding Insurance Coverage*

Trends in insurance coverage over time and by age have varied dramatically. As shown in Figure 1, insurance coverage rates among children under the age of 19 rose by roughly 15 percentage points between 1982 and 2007. In contrast, coverage rates among young adults (aged 19 to 29) remain as low as, and in some cases much lower than, they were 25 years ago. Indeed, Figure 2 shows that, over the past 25 years, insurance rates have diverged dramatically for older teenagers (aged 16–17)

[3] Workers who receive employer-provided health insurance could receive coverage of their children as a tax-free benefit. When children turn age 19 (or age 24 if in school), however, health insurance coverage through their parent's employer becomes a taxable benefit.

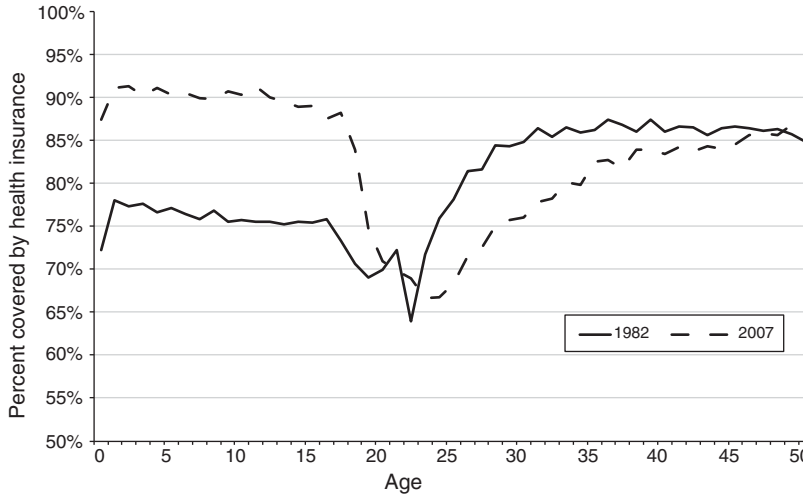

FIGURE 1. RATE OF HEALTH INSURANCE COVERAGE BY AGE, 1982 AND 2007

Source: Authors' calculations from the Current Population Survey.

FIGURE 2. HEALTH INSURANCE COVERAGE RATES OVER TIME

Source: Authors' calculations from the Current Population Survey.

and young adults (aged 20–21). Whereas the two groups had very similar insurance rates in 1982, there was a 17 percentage point gap by 2007. Fewer than 70 percent of these young adults (and fewer than two-thirds of 23 year olds—not shown here) are currently covered by health insurance. Because of low rates of insurance coverage among young adults, they comprise a large fraction of all those without health insurance. Indeed, one-third of all the uninsured in the US fall into this young adult age group. By the time individuals hit 30 and for the rest of their lives, a combination of public and private insurance again leads to relatively high rates of coverage.

B. *Legislative Background*

While Medicaid policy in the 1980s and early 1990s dramatically expanded coverage for pregnant women, infants, and young children, coverage expanded relatively little for teenagers during this time. Teenagers who received Aid to Families with Dependent Children (AFDC) were always covered by Medicaid, but the income eligibility limits for AFDC—and therefore Medicaid—coverage were extremely low in most states. For example, income eligibility limits for 18-year-olds were 15 percent of the federal poverty line in Alabama, 28 percent in Florida, 41 percent in New Jersey, 33 percent in Ohio, and 17 percent in Texas (Margo Rosenbach et al. 2003).

The only substantial Medicaid expansion that affected teenagers was the provision of OBRA 1990 requiring that all states provide Medicaid to children with incomes below the federal poverty line and birth dates after September 30, 1983 (US House of Representatives, Committee on Ways and Means 1994). This change led to a gradual expansion of Medicaid eligibility to older children and teenagers. For older teenagers, however, Medicaid eligibility remained extremely restrictive through 1997, when those born after September 30, 1983 were no more than age 14.

This changed in 1997 upon the introduction of SCHIP, which was created by the Balanced Budget Act of 1997 (BBA 97) and enacted in August of that year. This legislation provided federal matching funds for states to expand coverage up to age 19, and to children in families with incomes beyond previous Medicaid eligibility levels. States could either expand their existing Medicaid programs or create new programs. States launched their SCHIP programs at different times, but all did so within a fairly narrow window. Most began enrolling children in 1998 and by the end of 1999 all SCHIP programs were up and running (Judith Wooldridge et al. 2003). States were allowed to cover children in families up to 200 percent of the poverty line, but they could extend those income limits even further, depending upon the income thresholds in their existing Medicaid programs. This provision generates some variability in income eligibility limits across states, although the range is very narrow for most states.

Table 1 details the variation in the implementation and provisions of Medicaid expansions, including SCHIP programs, across states around the time that SCHIP was introduced. In most states, children lost coverage at age 14 in 1997 and at age 19 in 1999. There is some variation in the income eligibility limits for SCHIP across states, with limits ranging from 100 to 300 percent of the federal poverty line. However, the majority of states ultimately used an eligibility limit of 200 percent of the poverty line.

BBA 97 intentionally allowed states to incorporate features to minimize crowd-out into their SCHIP programs. For example, states are permitted to impose waiting periods before enrollment in SCHIP and most states have chosen to do so (see, Jonathan Gruber and Kosali Simon 2008). In addition, states are permitted to charge premiums and copayments for services, so long as total out-of-pocket costs (including premium) do not exceed 5 percent of family income. These features may be expected to reduce both take-up and crowd-out relative to rates that were observed in prior Medicaid expansions.

Table 1—Changes in Income and Age Eligibility Limits Associated with Medicaid Expansions and the Introduction of SCHIP

	Percent of federal poverty line				Upper age limit		
State	1995	1997	1999	2001	1995	1997	1999
Alabama	100	133	200	200	12	14	19
Alaska	100	100	200	200	12	14	19
Arizona	100	100	150	200	16	18	19
Arkansas	100	100	200	200	12	14	19
California	100	100	200	250	12	14	19
Colorado	100	100	185	185	12	14	19
Connecticut	185	185	300	300	12	14	19
Delaware	100	100	200	200	19	19	19
District of Columbia	100	100	200	200	12	14	19
Florida	100	100	200	200	12	19	19
Georgia	100	100	200	200	19	19	19
Hawaii	300	300	300	200	19	19	19
Idaho	100	100	150	150	12	14	19
Illinois	100	100	185	185	12	14	19
Indiana	100	100	150	200	12	19	19
Iowa	100	100	185	200	12	14	19
Kansas	100	100	200	200	12	16	19
Kentucky	100	100	100	200	12	19	19
Louisiana	100	100	133	200	12	14	19
Maine	125	125	185	200	19	19	19
Maryland	185	185	200	200	12	14	19
Massachusetts	200	100	200	200	12	14	19
Michigan	150	150	200	200	16	18	19
Minnesota	275	275	275	275	18	18	19
Mississippi	100	100	100	200	12	14	19
Missouri	100	100	300	300	18	18	19
Montana	100	100	100	150	12	14	18
Nebraska	100	100	185	185	12	14	19
Nevada	100	100	200	200	12	14	19
New Hampshire	185	185	300	300	19	19	19
New Jersey	100	100	200	350	12	14	19
New Mexico	100	185	235	235	12	19	19
New York	160	100	230	250	15	14	19
North Carolina	100	100	200	200	18	18	19
North Dakota	100	100	100	140	12	18	19
Ohio	100	100	150	200	12	14	19
Oklahoma	100	100	185	185	12	14	18
Oregon	100	100	170	170	12	19	19
Pennsylvania	185	185	235	235	12	14	19
Rhode Island	100	250	250	250	12	19	19
South Carolina	100	100	150	150	12	14	19
South Dakota	100	100	140	200	12	19	19
Tennessee	100	400	200	200	12	18	19
Texas	100	100	100	200	12	14	19
Utah	100	100	200	200	18	18	19
Vermont	225	225	300	300	18	18	18
Virginia	100	100	185	200	19	19	19
Washington	200	200	200	250	19	19	19
West Virginia	150	150	150	200	19	19	19
Wisconsin	100	100	100	185	12	14	19
Wyoming	100	100	100	133	12	14	19

Sources: Shore-Sheppard (2003), Centers for Medicare and Medicaid Services (2004), and National Governors Association (1995, 1997, 1999, and 2001).

Like Medicaid coverage, private coverage through a parent's policy has traditionally ended on a child's nineteenth birthday (or, for full-time students, the child's twenty-fourth birthday). Indeed, Michael Anderson, Carlos Dobkin, and Tal Gross

(2010) document a substantial discontinuity in private insurance coverage at exact age 19. These age limits for parental insurance coverage reflect the tax code, which specifies that the tax exclusion for health insurance includes coverage of a dependent child until their nineteenth (or twenty-fourth if enrolled in school) birthday (Department of the Treasury Internal Revenue Service 2009). While, in principle, an employer *could* have extended coverage to older children, the potentially complicated tax implications provided a very strong disincentive. Extended parental coverage laws have mandated that in spite of the tax implications, certain firms must extend coverage to young adult children.

Virtually all extended parental coverage laws have been implemented by individual states over the past five years. While there have been proposals to extend parental coverage at the federal level, all of the laws to date have been implemented by states. The first such law was implemented in Utah in 1994 where young Mormon adults were losing health insurance while carrying out their traditional two years of missionary work (Associated Press 2007). The state legislature responded with a policy allowing parents to cover these young adults until their twenty-sixth birthdays. Jennifer Kriss et al. (2008) provides an extensive description of the laws introduced through 2007, most of which were implemented in the few years preceding that date.

Table 2 summarizes these extended parental coverage laws. The table includes the years when extended parental coverage laws were implemented, the limiting age of dependency status (coverage expires at that birthday), as well as whether the young adult must be a student, must be unmarried, or cannot have their own dependents. The limiting age of dependency status, or age up to which young adults are covered, is 25 in the majority of states that passed such laws, with a few states opting for age 24, 26, or 30. Nearly all states require young adults to be unmarried. While most states allow nonstudents to qualify under extended parental coverage laws, a minority require the young adults to be enrolled in school.[4] A few states require young adults not to have any dependents of their own. As of the end of 2008, 25 states had implemented laws.[5]

Eligibility for extended parental coverage depends not only on the characteristics of the adult child, but also on the characteristics of the parents' insurance coverage. Naturally, parents who do not have private insurance coverage for themselves will not be able to extend coverage to their adult children. In addition, roughly 55 percent of those parents who have employer-provided health insurance are covered by firms that self-insure (William Pierron and Paul Fronstin 2008). Under the Employee Retirement Income Security Acts of 1974 (ERISA), self-insured firms are exempt from state-level health insurance regulation. The implication for extended parental coverage laws is that a substantial fraction of parents who have employer-provided health insurance do not have the option to extend it to their adult children under existing laws.

[4] Providing health insurance to students up until their twenty-fourth birthday is a nontaxable benefit even without these laws, but employers are not required to offer it. States that require school enrollment for extended parental coverage are expanding coverage by mandating the provision of this benefit as well as extending the age at which it is offered. In all of our empirical results presented below, we include these states as having an extended parental coverage law despite its potentially limited effect. We have also experimented with dropping these states and our results were virtually unchanged.

[5] See the notes to Table 2 for sources.

Table 2—Extended Parental Coverage Law Details

State	Year law was implemented	Limiting age of dependency	Must be a student	Must be unmarried	Cannot have dependents
Colorado	2006	25		X	
Delaware	2007	24		X	X
Florida	2007	25		X	
Idaho	2007	25	X	X	
Indiana	2007	24			
Maine	2007	25		X	X
Massachusetts	2006	25			
New Hampshire	2007	26		X	
New Jersey	2006	30		X	X
New Mexico	2005	25		X	
Rhode Island	2007	25	X	X	
South Dakota	2005	24	X		
Texas	2003	25		X	
Utah	1994	26		X	
Virginia	2007	25	X		
Washington	2007	25		X	
West Virginia	2007	25		X	

Notes: Entries in table show whether law was in effect as of December of that year. Age of dependency is birthday at which dependency expires.

Sources: Jessica Kronstadt, Safiya Mojerie, and Sonya Schwartz (2007); Kriss, et al. (2008); National Conference of State Legislatures (NCSL) (2008), and our own reading of state laws.

C. *Prior Literature*

Although the decision to remain uninsured may be rational if young adults are risk neutral, there are reasons to believe that some form of market failure may also be at play here. Helen Levy (2007) analyzes several reasons underlying the low and declining insurance rates among young adults. First, increasingly unstable employment patterns in this population, including frequent job changes and employment in part-time and temporary positions, diminish the likelihood of employer-sponsored health insurance coverage. Second, adverse selection, which has been widely documented in health insurance markets (see, David M. Cutler and Sarah J. Reber 1998), implies actuarially unfair premiums for young adults in both group and nongroup insurance markets, and therefore lower insurance rates. Increasing health care costs have likely exacerbated this actuarial unfairness for healthy, young adults. Both of these potential explanations for falling insurance rates suggest a potential role for the government. Even if the decision were rational for individual young adults, one could still argue that the government should intervene in this market if young adults' failure to purchase insurance imposes negative externalities on others. For instance, Anderson, Dobkin, and Gross (2008) find that young adults overuse expensive emergency room care because they lack health insurance, and those costs may be transferred to others.

Prior public interventions into insurance markets through Medicaid expansions have been studied extensively. However, that literature provides many conflicting estimates of take-up and crowd-out. For example, Cutler and Gruber (1996) identify the effects of Medicaid expansions using a measure of "simulated eligibility" that quantifies the legislative variation across states and years. They find crowd-out

rates of 30 to 50 percent.[6] John C. Ham and Lara Shore-Sheppard (2005), however, suggest that these results are sensitive to specification and choice of dataset. Other papers, using longitudinal datasets, have found lower, but nonzero, crowd-out rates (Kenneth Thorpe and Curtis Florence 1999; Ezel Yazici and Robert Kaestner 2000; Linda J. Blumberg, Lisa Dubay, and Stephen A. Norton 2000). Moreover, David Card and Shore-Sheppard (2004) use a regression discontinuity design to analyze expansions of Medicaid that affect those under 100 percent and 133 percent of the poverty line, and find that the coverage effect is low because take-up is low rather than because of large crowd out. On balance, the evidence on crowd-out in Medicaid expansions is quite mixed.

The more recent literature on SCHIP has also relied primarily on variation in program generosity across states and years to identify the effect on insurance coverage. Using this variation, Anthony T. LoSasso and Thomas C. Buchmueller (2004), for example, conclude that 9 percent of eligible children gained public insurance coverage, but that crowd-out was nearly 50 percent. Similarly, Gruber and Simon (2008) use the SIPP and find take-up rates ranging from 5 to 15 percent depending on the specification. Their central crowd-out estimate is 60 percent, although the estimates that are most closely related to the results presented in the current paper are in the 20 to 40 percent range. An exception is Lindsey Jeanne Leininger (2010), which uses the differential eligibility expansions for very low-income 15–17-year-olds, as compared to younger children in the same income range, after 1997. In a sample of 8 states, she finds a 28 percentage point increase in public insurance coverage and a 40 percent crowd-out rate.

Our contribution, relative to this body of previous literature, is a new identification strategy, which relies on the discontinuity in eligibility at age 19, to identify the impact of public insurance on coverage rates and crowd-out.[7] Because our identification strategy is so simple, we are able to show the basis for many of our results in graphs. In addition, our evidence focuses on the marginal older teenager (aged 16–18) who is induced to participate in public insurance due to the introduction of SCHIP rather than the marginal child (aged 0–18) who is induced to participate, as in the prior literature. Therefore, our findings are more relevant for thinking about the likely effects of expanding SCHIP coverage to older ages.

The literature on parental extended coverage laws is much less developed because the laws have been introduced so recently. To our knowledge, this paper is the first to address the impact of these laws on rates of insurance coverage.

II. Data

For our analysis, we use several years of data from the March Current Population Survey (CPS). The CPS is a monthly survey of about 50,000 households (although the sample size varies somewhat over time) conducted by the Bureau of Labor

[6] Crowd-out rates here, and throughout the paper, are defined as the decline in private insurance coverage as a percentage of the increase in public insurance coverage.

[7] Anderson, Dobkin, and Gross (2010) also use the discontinuity in insurance coverage at exact age 19 to identify the impact of insurance coverage on the use of emergency rooms, outpatient and inpatient services.

Statistics. Each month, information about employment status and demographic characteristics is obtained from individuals aged 16 and older. The March supplement to the CPS asks additional questions, including whether individuals had private health insurance, public health insurance, or were uninsured (in more recent years) in the preceding calendar year.

Our analysis of SCHIP uses these data between the years 1992 and 2009, whereas our analysis of extended parental coverage laws uses the data between 2000 and 2009. These datasets provide reports of insurance coverage in the periods of 1991 to 2008 and 1999 to 2008, respectively. We use different sample windows for these exercises to focus more closely on the periods in which the laws changed.

Several potential concerns must be addressed when using the CPS health insurance variables. First, the CPS was redesigned in 1995, which led to some changes in the reporting of health insurance. In addition, new health insurance verification questions were added in 2001 which, again, affected the reporting of health insurance coverage. These changes should not contaminate our analysis, however, as long as the reporting changes did not vary discretely with age, which seems unlikely.[8] Second, there is a long-standing concern that CPS respondents may actually provide their current insurance status when asked about their insurance status in the prior year. Thus, health insurance variables are potentially measured with error (Katherine Swartz 1986). The quasi-experimental nature of our identification strategy, however, is unlikely to be seriously affected by this problem. All pre- and post-policy change years are grouped together so these errors are likely to generate only a relatively small number of miscodings. Third, the recent literature on SCHIP has raised the concern that, as states outsource their public insurance plans to HMOs, some individuals who are actually on state Medicaid and SCHIP programs may incorrectly report that they are on nongroup private health insurance. Such incorrect reporting could bias downward estimates of both take-up and crowd-out. Indeed, Lo Sasso and Buchmueller (2004) find that their results are sensitive to the treatment of reported nongroup private insurance coverage. We will also examine the sensitivity of our results to this issue.

Finally, the literature on Medicaid and SCHIP has raised questions about how to code individuals who report both public and private health insurance coverage in the prior year. If these individuals were on private health insurance at the beginning of the year and dropped it in order to take up public insurance, then coding them as being on both types of insurance coverage would bias downward estimates of crowd-out. Gruber and Simon (2008) explored this issue using panel data in the SIPP and found that 40 percent of children who reported both public and private insurance in the middle wave also reported overlapping public and private coverage before and after the middle wave. This finding suggests that it is, in many cases, appropriate to code these individuals as holding both private and public insurance. Nonetheless, we explore the sensitivity of our estimates to this concern below.

One other important aspect of the data for our purposes is the way that children's incomes are measured.[9] The CPS is a household survey, so a child's family income is the same as his or her parents as long as they are living in the same household. Children

[8] We have verified that the results of our analysis are not sensitive to excluding the data collected prior to 1995.
[9] See http://www.census.gov/population/www/cps/cpsdef.html for more detail.

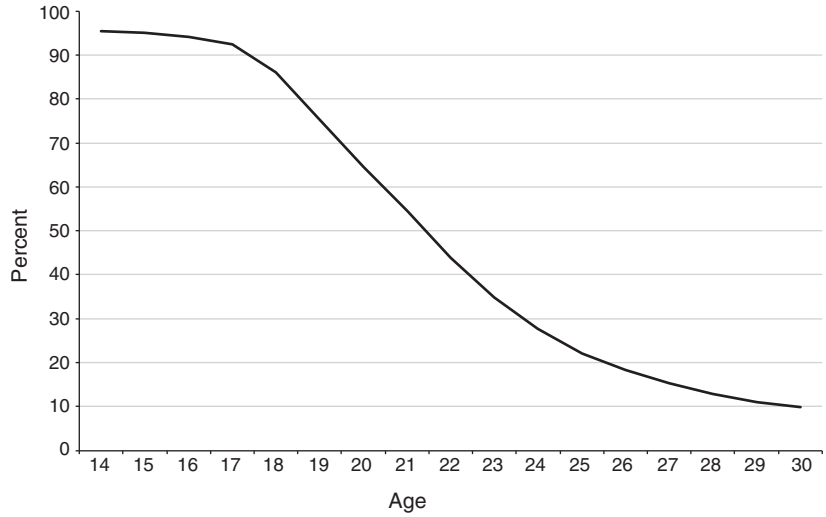

Figure 3. Percentage of Young Adults Who Live With Their Parents

Source: Authors' calculations from the Current Population Survey.

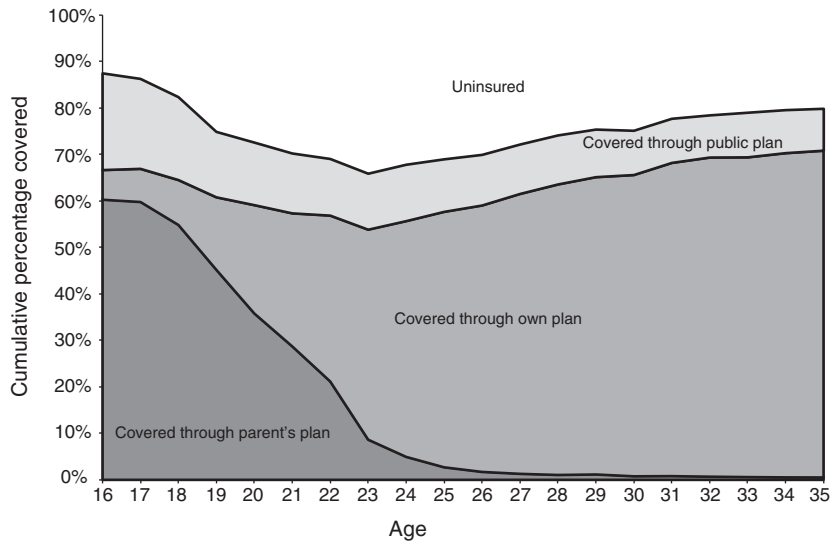

Figure 4. Health Insurance Coverage Status: 2000–2008, by Age

away at school are also assigned to their parents' household, so family income is assigned from the parent. Children who have moved out of their parents' house and are not in school would be categorized as living independently and would report their own income, not their parents. Not surprisingly, we find that the number of children who still "live" with their parents (or are away at school) changes very rapidly over the age ranges we consider in this paper, as shown in Figure 3. This is an important aspect of the data because some of our analyses will distinguish individuals by their income. We will return to the specifics of this issue as appropriate later in the paper.

To capture the nature of the data available to us, we present Figure 4, which displays the age pattern of health insurance status, pooling data from the 2000 to 2008 period. This figure is comparable to figures reported by Levy (2007). It shows that the majority of teens are covered by a parent's health insurance plan through age 18, and then parental coverage drops off sharply through the early 20s. Coverage from a public plan through age 18 captures a substantial share of those without parental coverage, but that drops off as well soon after that. Own private coverage, which includes spousal coverage, makes up a lot of the lost coverage through the late teens and early 20s, but not enough to prevent a sizeable drop-off in insurance coverage. Rates of coverage by any insurance rebound in the middle 20s, but even by age 35 do not reach the level of childhood coverage. Insurance coverage status measures for each individual in the relevant CPS surveys provide the dependent variable in our subsequent econometric analyses.

III. The Impact of SCHIP

A. *Identification Strategy*

Our empirical strategy for identifying the effect of SCHIP on older teenagers relies on the age discontinuity in eligibility. As shown in Table 1, 36 states (including almost all of the large states—NY, CA, TX, IL, for instance) raised the age limit of eligibility to age 19 between 1997 and 1999 in the immediate aftermath of the introduction of SCHIP. We take advantage of this, simplifying our analysis by focusing on the difference in insurance coverage before and after 1998 for those who are over and under age 19. Our approach is intended to determine whether an increase in insurance coverage occurred beginning in 1999 for those who are under age 19 relative to those who are aged 19 and over.

We recognize that there are alternative ways to identify a policy effect in this context. For instance, we could create a state- and year-varying indicator for eligibility at age 19 that would capture the fact that some states increased their upper age limit to this level earlier than the introduction of SCHIP. Alternatively, we could also take advantage of the varying income thresholds that exist across states and generate a simulated percentage of individuals who would be covered by health insurance in a state and year at a given age. This is the approach used by Janet Currie and Gruber (1996) and others. We have estimated models using these alternative approaches and obtained results that are very similar to the results from our analysis of the age 19 discontinuity in terms of their behavioral implications. We report our results using this simplified model, recognizing that it does not take advantage of every last bit of variation that exists in the data, because it yields results with the same meaning and it is much more transparent, as seen in the following section.

B. *Descriptive Analysis*

An advantage of our identification strategy is that any effects should be apparent in a simple, graphical analysis. Indeed, Figure 5A provides suggestive evidence that an age discontinuity in insurance coverage emerged after BBA 97. The figure shows

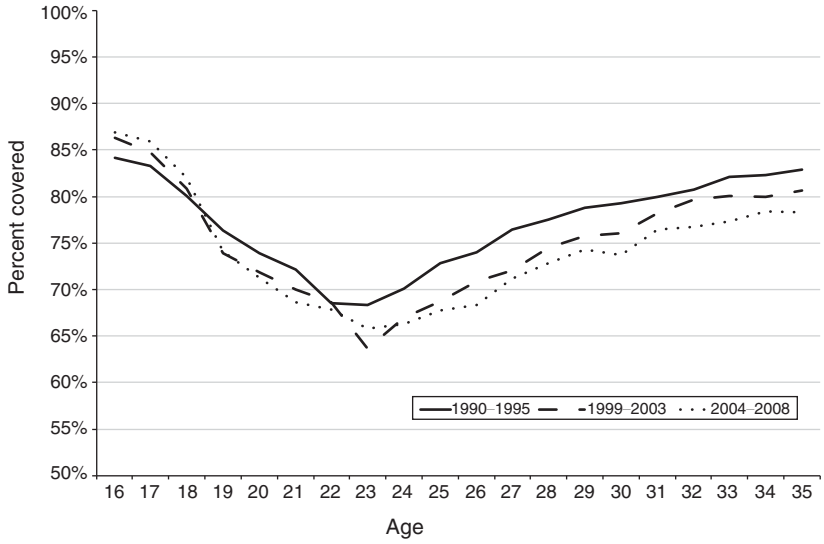

Figure 5A. Percent Covered by Insurance

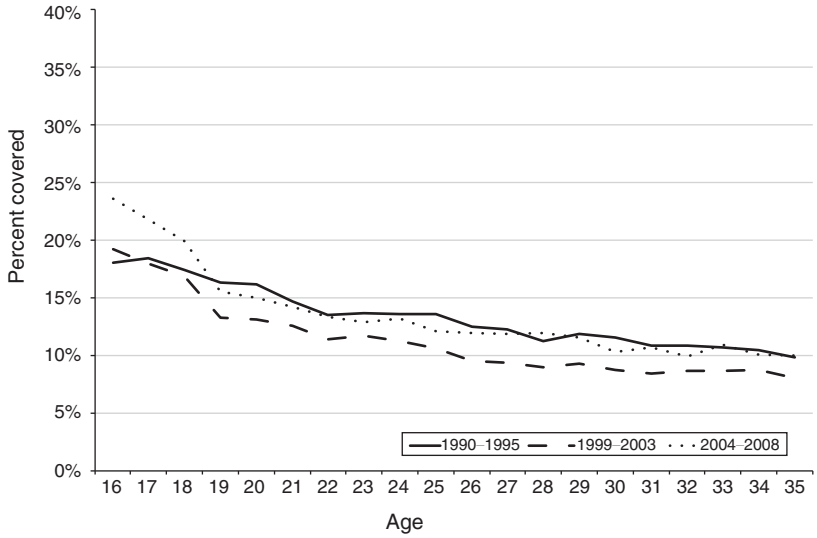

Figure 5B. Percent Covered by Public Health Insurance

the age pattern of insurance coverage in three different time periods: 1990–1995 (before SCHIP), 1999–2003 (immediately after SCHIP), and 2004–2007 (several years after SCHIP). At most ages, the profile of insurance coverage falls relatively uniformly over time. The exception to this pattern is an *increase* in insurance coverage for individuals under 19 years of age in the post-SCHIP periods relative to the pre-SCHIP time period. This suggests that the introduction of SCHIP generated a substantive increase in insurance coverage for older teenagers.

Public insurance coverage displays a similar pattern, as shown in Figure 5B. The age profile of coverage shows that the percentage of individuals covered by public

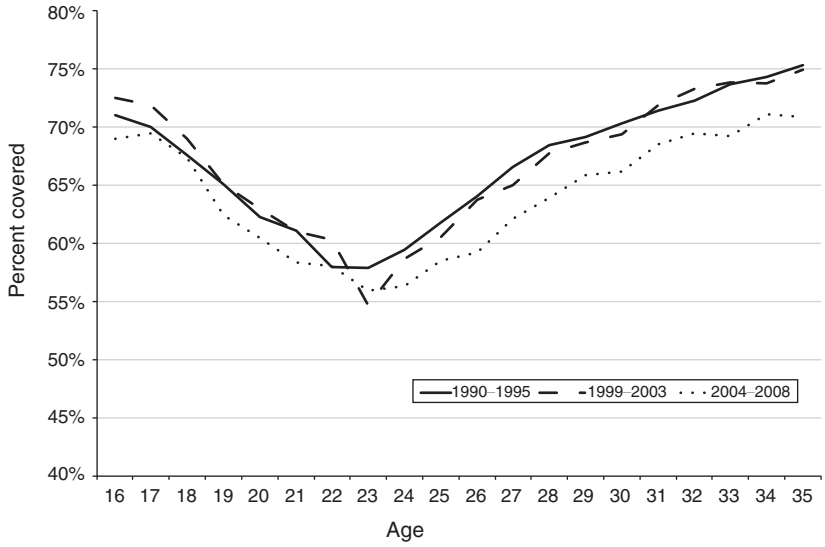

FIGURE 5C. PERCENT COVERED BY PRIVATE HEALTH INSURANCE

insurance fell over time at most ages with the notable exception of 16–18-year-olds, who experience a relative increase in public insurance coverage in the post-SCHIP periods. Figure 5C provides the same graphical analysis for private insurance coverage. If crowd-out were a substantial issue for this age group, we would expect to see a relative decline in private insurance coverage for 16–18-year-olds after SCHIP was introduced. Interestingly, this figure suggests that there was, if anything, an increase in private insurance coverage. This figure provides suggestive evidence that the magnitude of crowd-out in this age group is unlikely to be large.

C. *Econometric Specification*

Our regression analysis formalizes this graphical analysis, using the following framework:

$$(1) \qquad Insurance_{iast} = \alpha + \beta Below19_a \cdot Post1998_t + \delta_a + \tau_t + \pi_s + \rho Unemp_{st} + X_{iast}\theta + \varepsilon_{iast},$$

where $Insurance_{iast}$ is a measure of insurance status for individual i in age group a, state s, and year t. Our independent variable of interest is an indicator for being under the age of 19 after 1998.[10] The regression includes a full set of age, year, and state fixed effects. In addition, we control for differences in economic conditions

[10] In our analysis, we assume that the age reported in the March CPS corresponds to the individual's age in the prior year, the year for which health insurance variables are reported. However, we have confirmed that our results are not sensitive to excluding 19-year-olds—who might plausibly have been below 19 and therefore in the treatment group, rather than the control group, during the prior year.

Table 3—The Effect of SCHIP on Insurance Coverage

	Percent covered by any insurance (1)	Estimation results for dependent variable: Any insurance coverage (2)	Public insurance coverage (3)	Private insurance coverage (4)
Full sample	76.1	0.032**	0.045**	−0.004
$N = 291{,}118$		(0.005)	(0.005)	(0.004)
Sample living with parents	81.8	0.029**	0.037**	0.001
$N = 218{,}818$		(0.010)	(0.005)	(0.008)
By income				
< 150 percent of poverty line	63.9	0.067**	0.084**	−0.004
$N = 41{,}168$		(0.016)	(0.016)	(0.006)
150–300 percent of poverty line	80.6	0.043**	0.058**	0.001
$N = 77{,}562$	(0.019)	(0.004)	(0.018)	
> 300 percent of poverty line	91.9	0.002	0.005**	−0.002
$N = 100{,}088$		(0.005)	(0.001)	(0.006)

Notes: Each cell in columns 2–4 presents the results from separate difference-in-difference regressions models. Control variables include state, year, and age fixed effects, the state- and year-specific unemployment rate, and individual-level covariates. Standard errors are clustered on age. The sample includes all observations between the ages of 16 and 22, using the 1992–2009 CPS.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

across states, using the state- and year-specific unemployment rate.[11] We also control for individual-level covariates, including gender, marital status, student status, an indicator for residence with a parent, household income as a share of the poverty line, and the square of household income as a share of the poverty line.[12] We use OLS to estimate the regressions and cluster our standard errors at the age level.[13] We limit the sample to ages 16 to 22, so we are including only three years above and below the age cutoff. The implicit identifying assumption of our regression is that, in the absence of SCHIP, there would not have been differential trends in insurance coverage at ages around the age 19 eligibility limit.

D. *Results*

Table 3 shows our main results for SCHIP using 1992–2009 March CPS data.[14] Column 1 in this table displays the percentage of individuals in the relevant population subgroup that was covered by any form of insurance prior to policy implementation. This is a useful metric for gauging the magnitude of the estimated effects reported in the remainder of the table. Each cell in columns 2–4 reports the results

[11] Results are similar when we also include state-specific linear trends in our econometric specification.

[12] Our results are insensitive to including income as a control variable.

[13] Because we have only seven clusters, we verified that the statistical significance of our main results is insensitive to use of the wild cluster bootstrap method for clustered standard errors outlined in A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller (2008).

[14] All reported regressions are based on models that do not use CPS sampling weights. We have estimated all models both with and without weights and obtained similar results.

from a different regression; the reported coefficient reflects the estimated value of β from equation (1), which is the coefficient on an indicator for being under the age of 19 after 1998. Each column shows coefficients and standard errors for a different dependent variable, and the rows show results for different subsamples.

The first row of Table 3 shows results for the full sample (both those living with parents and independently) of 16–22 year olds. Our results suggest a statistically significant 3.2 percentage point increase in insurance coverage for older teenagers associated with the introduction of SCHIP (column 2). This reflects a 13 percent reduction in the rate of uninsurance. The third column shows a larger, statistically significant 4.5 percentage point increase in public insurance, and the fourth column shows a statistically insignificant 0.4 percentage point decrease in private insurance coverage.[15] The implied level of crowd-out is 9 percent (0.004/0.045), although this estimate is not significantly different from zero. While this point estimate is lower than those in the rest of the SCHIP literature, the 95 percent confidence interval includes crowd-out rates as high as about 30 percent.[16]

The lower rows of Table 3 explore heterogeneity in the effect of SCHIP by income category. Because we prefer a measure of household income that does not reflect the temporarily low incomes for otherwise middle- and high-income young adults who are transitioning into the labor market, we restrict this analysis to young adults who live with their parents.[17] Therefore, the second row replicates our main results for the full sample of 16–22 year olds who live with their parents. The increase in insurance coverage is somewhat smaller than the results in the full sample, but the difference is small and not statistically significant.

For the lowest income group, with household incomes below 150 percent of the poverty line, we see a substantial, 6.7 percentage point increase in insurance coverage. This estimated impact of SCHIP suggests that it reduced uninsurance rates by 19 percent for this group. Estimates of crowd-out in this lowest income category are small. The 8.4 percentage point increase in public insurance is accompanied by a statistically insignificant 0.4 percentage point decline in private insurance coverage, suggesting a 5 percent crowd-out rate.

For the middle income group, with household incomes between 150 percent and 300 percent of the poverty line, take-up rates are lower, which is not surprising, and the overall increase in insurance coverage is smaller, a 4.3 percentage point increase compared to a 6.7 percentage point increase for the lower income group. As a percentage of the uninsured group, the impact is sizeable. In this group, 19.4 percent are uninsured, so a 4.3 percentage point reduction amounts to a decline of 22 percent. Crowd-out for this income group is wrong-signed and statistically insignificant. For

[15] The coefficients from the public and private insurance models would sum to that for any insurance model if the two forms of insurance were mutually exclusive. The nature of the questions asked in the CPS, however, makes it possible for a respondent to report both types of coverage in the past year. We discuss this issue in more detail below.

[16] We use the delta method to calculate the standard deviation of the ratio of two random variables. Here, and in the remainder of the table, a standard deviation is about 10 percentage points.

[17] We have estimated regression models of the same form as equation (1), but replacing the insurance indicator with a "living with parents" indicator. The coefficient on below 19 after 1998 is not statistically significant in this specification (point estimate equals 0.4 percent with a standard error of 0.5 percent). We also estimated our models including data on children living independently, and we obtained similar results.

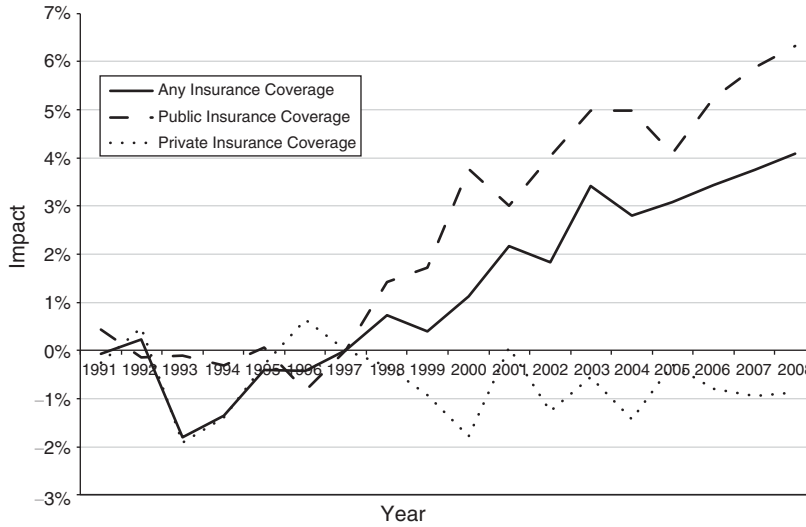

Figure 6. Impact of SCHIP on Health Insurance Coverage, by Year

Table 4—The Effect of SCHIP on Insurance Coverage
Dynamic Specification

	Estimation results for dependent variable		
	Any insurance coverage (1)	Public insurance coverage (2)	Private insurance coverage (3)
Under age 19			
* Years 1991–1993	−0.005	0.001	−0.005
	(0.009)	(0.006)	(0.007)
* Years 1994–1996	−0.007**	−0.004	−0.004
	(0.002)	(0.006)	(0.005)
* Years 1998–1999	0.006	0.016**	−0.006
	(0.004)	(0.004)	(0.004)
* Years 2000–2002	0.018**	0.036**	−0.009
	(0.003)	(0.004)	(0.005)
* Years 2003–2005	0.031**	0.047**	−0.007
	(0.005)	(0.007)	(0.006)
* Years 2006–2008	0.038**	0.058**	−0.009
	(0.001)	(0.005)	(0.005)

Notes: Each column presents the results from separate difference-in-difference regressions models. The omitted period is 1997. Control variables include state, year, and age fixed effects, the state- and year-specific unemployment rate, and individual-level covariates. Standard errors are clustered on state. The sample includes all observations between the ages of 16 and 22, using the 1992–2009 CPS. $N = 291{,}118$.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

the highest income group, with household incomes above 300 percent of the poverty line, we expect little impact of the introduction of SCHIP, because very few states extended eligibility to teenagers in this part of the income distribution. Indeed, all coefficients are very small and they are generally statistically insignificant.

In Figure 6 and Table 4, we examine the time pattern of the change in insurance coverage patterns for individuals under the age of 19, to ensure that it corresponds to the timing of SCHIP implementation. In Figure 6, we display the coefficients on year-by-year interactions with an indicator for being under the age of 19 for public, private, and any health insurance coverage. In Table 4, we repeat the same analytical framework, but we use six separate independent variables of interest, which interact the dummy variable for being under the age of 19 with dummies for different time periods. The figure provides more detail regarding the time series patterns whereas the table makes it easier to report exact estimates and to conduct statistical inference. In both cases, the omitted time period is 1997, the year before changes in insurance coverage should have begun, so all coefficients are measured relative to 1997. Each column of the table reports the six key coefficients from a different regression.

Taking these results collectively, our analysis supports the conclusion that the introduction of SCHIP had a causal impact on health insurance coverage for those under the age of 19.[18] In Figure 6, we see that public health insurance coverage for those under the age of 19 was roughly constant prior to 1997 and then rose dramatically over the course of the following decade. No such trend is observed for private health insurance coverage, so any insurance coverage closely tracks the pattern for public coverage. These patterns are confirmed in Table 4. The lack of any pre-existing differential trends and the growth in public insurance coverage after 1997 for those under the age of 19 supports our causal interpretation of the results. The increase in the impact on public insurance coverage through our sample period is also consistent with overall SCHIP enrollment patterns, which nearly doubled between 2000 and 2007 (Vernon Smith et al. 2008), particularly through 2003.[19] Our estimates for private insurance coverage, reported in the third column, provide only statistically insignificant evidence of crowd-out.

E. *Alternative Measures of Crowd-Out*

The prior literature on crowd-out suggests two alternative ways to measure insurance status, which could potentially increase our estimate of crowd-out. One alternative is to consider any increase in nongroup private insurance to be an increase in public insurance coverage. This alternative measure of public insurance addresses the concern that CPS respondents who are on SCHIP programs might erroneously report that they had private insurance coverage. Crowd-out, in this case, should be measured as the reduction in group insurance coverage, as a percentage of the increase in public and nongroup insurance coverage. When we follow this approach, our estimates of crowd-out rise. Specifically, we find a 5.5 percentage point increase

[18] We also estimated models completely analogous to those reported in Tables 3 and 4, but for a group of 19 to 22 year olds who would not be eligible for SCHIP coverage, and tested for a discontinuity in insurance coverage at age 21. Virtually all coefficients in these models are wrong-signed and statistically insignificant, supporting the notion that our results are not capturing differences in underlying trends. The results are reported in the online Appendix.

[19] States used different strategies to boost enrollment, including outreach efforts, community-based application assistance, simplified application and renewal procedures, simplified eligibility rules, reduced premiums, and others (Kirsten Wysen, Cynthia Pernice, and Trish Riley 2003). These efforts continued through the early 2000s and led to rapidly rising enrollments, particularly through 2003.

in public plus nongroup insurance, and an insignificant 1.5 percentage point decline in group insurance coverage, for an overall crowd-out rate of 27 percent. However, crowd-out among the lowest income group—those with household incomes below 150 percent of the poverty line—remains low at only 9 percent.

Another alternative approach is to measure crowd-out as the reduction in reporting *only* private insurance coverage, as a percentage of the increase in public insurance coverage. This approach addresses the concern that individuals who are in the process of switching from private to public health insurance coverage may report both types of coverage. This approach is the most conservative possible solution to this problem because it assumes that all individuals who report overlapping public and private coverage are being crowded out. As mentioned above, Gruber and Simon's (2008) investigation of this issue in the SIPP suggests that this approach is likely too conservative, since roughly half of the individuals who report having both public and private insurance continue to do so in later interviews. Nonetheless, we used this approach and found a statistically significant decline in private insurance coverage, resulting in an overall crowd-out rate of 29 percent. Among the lowest income groups, the crowd-out rate is 21 percent and remains statistically insignificant. We consider these estimates to be upper bounds on crowd-out among older teenagers due to SCHIP expansions.

While our estimates of crowd-out are lower than those in the prior literature, we do not believe that they are necessarily inconsistent with the prior literature. Our identification strategy relies on a marginal SCHIP participant who is a 16- to 18-year-old who was made eligible for public insurance by SCHIP's expansion of the age eligibility limit. In contrast, the identification strategy in Gruber and Simon (2008) and Lo Sasso and Buchmueller (2004) relies on differences in SCHIP eligibility across all age groups and states. Their strategy, therefore, relies on a marginal SCHIP participant who is likely to be much younger and with higher income. For example, prior to the introduction of SCHIP, the minimum income eligibility limit for 1–5-year-olds was 133 percent of the federal poverty line, whereas the typical 18 year old had to meet the dramatically lower AFDC eligibility rules. So, if higher income individuals have lower take-up rates and higher crowd-out rates, it may not be surprising that our analysis estimates lower crowd-out.[20] This interpretation would be consistent with the existing literature that typically finds less crowd-out in the earlier Medicaid expansions which affected relatively low-income children (see e.g., Shore-Sheppard 2008) than in the later Medicaid expansions such as SCHIP, which affected higher income children (see e.g., Gruber and Simon 2008 and LoSasso and Buchmueller 2004). For the purposes of predicting crowd-out rates for further expansions of public insurance to older age groups—including recent proposals to expand SCHIP eligibility to adults in their early 20s—our lower estimates of crowd-out in SCHIP are likely more relevant than estimates from the prior SCHIP literature.

[20] We have also investigated the role that the older ages of the marginal children in our analysis relative to that in the earlier literature may play in the extent of crowd out. We found that the percentage of children's parents who are covered by group health insurance rises from around 58 percent at birth to almost 70 percent by age 18. This suggests that the potential for crowd out is greater for older children. This indicates that our finding of less crowd out relies more heavily on the lower incomes of the marginal children in our analysis.

IV. The Impact of Extended Parental Coverage Laws

A. *Identification Strategy and Econometric Specification*

The second portion of our analysis focuses on the impact of extended parental coverage laws. As shown in Table 2, there is substantial variation across states in the implementation of these laws. Our basic identification strategy relies on standard quasi-experimental variation comparing changes in outcomes in states that adopt laws before and after their implementation relative to changes in outcomes in states that did not introduce these laws. In addition, since the provisions of these laws in states that adopted them made certain population subgroups eligible and others ineligible, we also adopt triple-difference estimation strategies that provide even stronger methods of drawing causal conclusions.

Our basic regression framework is as follows:

$$(2) \quad Insurance_{iast} = \alpha + \beta Law_{st} + \delta_a + \tau_t + \pi_s + \rho Unemp_{st} + X_{iast}\theta + \varepsilon_{iast},$$

where $Insurance_{iast}$ is a measure of insurance status for individual i in age group a, state s, and year t. Our independent variable of interest is an indicator for living in a state that has adopted an extended parental coverage law in a year after the law was implemented. As in the SCHIP analysis, the regressions include a full set of age, year, and state fixed effects. We again control for differences in state-specific economic shocks using the state- and year-specific unemployment rate, and we control for individual-level covariates, including gender, marital status, student status, an indicator for residence with a parent, household income as a share of the poverty line, and the square of household income as a share of the poverty line. We continue to use OLS to estimate the regressions, and we cluster our standard errors at the state level.

To begin our analysis, we use a sample of all young adults (both those living with parents and independently) between the ages of 19 and 24, the age group that would be affected by laws, to estimate the aggregate impact of the law using standard difference-in-difference methods.[21] We then separate that sample into population subgroups that we would expect to be more and less affected by its introduction. Initially, we restrict the sample to include just those who meet the eligibility criteria in their state (as defined in Table 2).[22] For states that have not passed extended parental coverage laws, and therefore have not established eligibility criteria, we assume that the only eligibility criterion is that the young adult must be unmarried.

As an additional test, we run a triple-difference specification that uses the ineligible young adults in the states as the third difference. This approach can adjust for other within-state differences that may affect levels of health insurance coverage in the state. Since marital status is the main eligibility distinction, we are comparing,

[21] We exclude all data from Massachusetts in our analysis, because of the broader health insurance reform that was implemented at the same time as the extended parental coverage law.

[22] In different states, eligibility is determined by school enrollment, marital status, and number of children, all of which are readily identifiable in the CPS.

Table 5—The Effect of Extended Parental Coverage Laws on Insurance Coverage

	Percent covered by any insurance (1)	Estimation results for dependent variable: Any insurance coverage (2)	Public insurance coverage (3)	Private insurance coverage (4)
Panel A. Full sample				
Difference-in-difference	69.3	−0.0005	−0.013**	0.010
Full sample		(0.007)	(0.006)	(0.007)
$N = 127{,}106$				
Eligible sample	69.4	0.008	−0.014**	0.022**
$N = 100{,}218$		(0.006)	(0.006)	(0.007)
Triple difference				
Full sample	69.3	0.033**	−0.005	0.044**
$N = 127{,}106$		(0.013)	(0.012)	(0.010)
Panel B. Non-students				
Difference-in-difference				
Full sample	63.2	−0.001	−0.008	0.006
$N = 77{,}861$		(0.008)	(0.009)	(0.010)
Eligible sample	60.8	0.018**	−0.006	0.026**
$N = 54{,}417$		(0.008)	(0.009)	(0.011)
Triple difference				
Full sample	63.2	0.051**	0.005	0.054**
$N = 77{,}861$		(0.015)	(0.010)	(0.014)

Notes: Each cell in columns 2–4 presents the results from separate difference-in-difference regressions models. Control variables include state, year, and age fixed effects, the state- and year-specific unemployment rate, and individual-level covariates. Standard errors are clustered on state. The sample includes all observations between the ages of 19 and 24, using the 2001–2009 CPS.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

in essence, outcomes for married and unmarried young adults. Level differences in health insurance coverage between these two groups are likely, but our identifying assumption is that the timing of any differential changes in coverage is unrelated to the introduction of extended parental coverage laws. This identification strategy is compromised if unmarried young adults happened to receive greater health insurance coverage than married young adults in those states that implemented extended parental coverage laws in the years following implementation. We believe that this would be an unlikely random occurrence. We continue our analysis by further separating the sample into narrower population subgroups, who we might expect to be more and less likely to be affected by extended parental coverage laws.

B. *Results*

Tables 5 and 6 present the results of our analysis.[23] As in Table 3, the first column of these tables displays the percentage of individuals in the relevant population subgroups who are covered by any form of insurance. In columns 2–4 of both tables,

[23] All reported regressions are based on models that do not use CPS sampling weights. We have estimated all models both with and without weights and obtained similar results.

Table 6—The Effect of Extended Parental Coverage Laws on Insurance Coverage Among Targeted Populations

	Percent covered by any insurance (1)	Estimation results for dependent variable: Any insurance coverage (2)	Public insurance coverage (3)	Private insurance coverage (4)
All living with parents	74.5	0.016**	−0.013**	0.029**
$N = 60{,}550$		(0.007)	(0.007)	(0.007)
By income				
<150 percent of poverty line	51.2	−0.002	−0.020	0.001
$N = 8{,}624$		(0.028)	(0.029)	(0.023)
150–300 percent of poverty line	67.0	0.027**	−0.013	0.051**
$N = 19{,}847$		(0.012)	(0.011)	(0.014)
>300 percent of poverty line	85.0	0.007	−0.005	0.011
$N = 32{,}079$		(0.009)	(0.006)	(0.010)
By parents' insurance				
Group coverage	83.7	0.012	−0.014**	0.022**
$N = 44{,}221$		(0.007)	(0.005)	(0.007)
And firm size <100	81.0	0.036*	−0.032**	0.055**
$N = 10{,}264$		(0.020)	(0.010)	(0.020)
And firm size >100	85.3	0.004	−0.009	0.012*
$N = 33{,}957$		(0.008)	(0.005)	(0.007)
No group coverage	48.2	0.009	−0.001	0.018
$N = 16{,}329$		(0.020)	(0.018)	(0.011)

Notes: Each cell in columns 2–4 presents the results from separate difference-in-difference regressions models. Control variables include state, year, and age fixed effects, the state- and year-specific unemployment rate, and individual-level covariates. Standard errors are clustered on state. The sample includes observations on all targeted groups between the ages of 19 and 24, using the 2001–2009 CPS. All targeted groups are restricted to those who live with their parents and are eligible for coverage in treatment states and are unmarried in control states.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

every cell represents the results from a different regression. Different dependent variables are shown in different columns, and different samples and specifications are shown in different rows.

The first row of Table 5 shows the difference-in-difference results for the full sample. We do not find that extended parental coverage laws are associated with a statistically significant increase in insurance coverage overall or in private insurance coverage in this model. In the second row, we focus more narrowly on those young adults who would be eligible for this form of coverage. Perhaps not surprisingly, we see larger effects for this group; private insurance coverage is estimated to increase by a statistically significant 2.2 percentage points. The magnitude of the effect on private insurance coverage is larger in the triple-difference specification in the third row, at 4.4 percentage points.

Interestingly, the net effect on overall insurance coverage in both specifications is muted by an offsetting decline in public insurance coverage, which is statistically significant in the difference-in-difference specification for the eligible sample. This finding suggests the possibility of "reverse crowd-out" resulting from extended parental coverage laws. When young adults are able to move onto their parents'

private insurance plans, some of them appear to drop public insurance to do so. This reverse crowd-out might be expected to occur if the private insurance plan provides more generous coverage or better access to providers.[24]

Overall, our best estimate of the aggregate impact of extended parental coverage laws comes from our triple difference estimate of the effect on any insurance coverage. There we see that implementing these laws increases insurance coverage by 3.3 percentage points. With an underlying rate of uninsurance of 31 percent, this means that if every state implemented one of these laws, then uninsurance in this age group would fall by 11 percent.

In the second panel of Table 5, we restrict our analysis to individuals who are not students. Since the tax incentive for employer-provided health insurance has traditionally extended to students up to the age of 24, the impact of extended parental coverage laws should be smaller for students than for nonstudents. In the bottom of Table 5, we show results for nonstudents who should be more strongly affected.[25] The point estimates we report show that, as we would expect, increases in private insurance coverage associated with extended parental coverage laws are concentrated among nonstudents. The triple difference estimate for the sample of nonstudents indicates that these laws increased private insurance coverage by 5.4 percent, compared to 4.4 percent for the full sample. Although point estimates are consistent with our hypotheses, standard errors are sufficiently large that we cannot statistically reject that these differences are attributable to random variation. Panel B also shows that the decreases in public insurance coverage are not statistically significant. Although the additional evidence that is presented below is consistent with reverse crowd-out, the findings in panel B of Table 5 lead us to consider our overall evidence for reverse crowd-out to be suggestive, rather than conclusive.

Table 6 reports the results of additional difference-in-difference models by more narrowly defined population subgroups. To identify these subgroups, we need identifying information in the data that is only available on the parents' record, so we are required to restrict the population to those respondents living with their parents. This sample restriction has some important implications regarding selection that we explore more fully below, but we continue this discussion ignoring these issues for now. Because the main reason a young adult would be ineligible for extended parental coverage is marriage and few married young adults live with their parents, the triple-difference estimating strategy does not make sense in this context. In this part of the analysis, we restrict our attention to difference-in-difference models relying just on those eligible for extended parental coverage, as described earlier.

The first row of this table reports the results of estimates comparable to the second row of Table 5 (eligible sample) except that it focuses on just those young adults living with their parents. The results from this specification are somewhat

[24] Among our sample of 19 to 24 year-olds who are on Medicaid and live with their parents, 39 percent have parents with group health insurance. This fact suggests that there is scope for reverse crowd-out. One mechanism for such crowd-out is that young adults who were eligible for Medicaid due to a disability moved onto their parents' group health insurance after the passage of extended parental coverage laws. Indeed, some states simultaneously extended parental coverage both to young adult children and to disabled children of any age.

[25] We do not show results for students because there are so very few ineligible students that our triple-difference empirical strategy is far less compelling for this subsample.

larger than those in Table 5. The next rows of the table show effects by income category. Increases in private insurance are statistically insignificant for young adults in households with incomes below 150 percent of the poverty line. This finding is not surprising, since the parents in these households are less likely to have private health insurance to extend to their adult children.[26] The impact of the extended parental coverage is concentrated almost entirely among the group with incomes between 150 and 300 percent of the poverty line. Interestingly, there is not a statistically significant increase in insurance coverage for those with incomes over 300 percent of the poverty line. In the pre-policy period, though, 85 percent of these young adults had private insurance, so there was less scope for expanding insurance coverage in this income group than in the lower income groups.

In the bottom rows of Table 6, we examine heterogeneity in the effects by parents' insurance coverage. We expect the impact of extended parental coverage laws to be strongly concentrated among the children of parents who are covered by group health insurance plans. Although some states included nongroup insurance policies in their extended parental coverage laws, most focused on group health insurance policies. Moreover, since group health insurance policies are typically much less expensive than nongroup health insurance policies, young adults who become eligible for extended parental coverage have a much stronger incentive to take advantage of it if their parents have group insurance. In fact, we see that the effects of the extended parental coverage laws—including increases in private insurance and decreases in public insurance—are concentrated among those whose parents have group insurance coverage.[27]

In addition, among children whose parents have group health insurance, we expect the effects of extended parental coverage laws to be strongest for those whose parents work for relatively small firms. This prediction arises from the ERISA legislation of 1974, which exempts the insurance policies of self-insured firms from state regulation. As a result, 55 percent of those who have employer-provided health insurance have insurance policies that are not affected by these legal changes.[28] While we cannot observe in the CPS which parents work in self-insured firms, we do know that the probability of self-insurance increases dramatically with firm size. In firms with fewer than 200 employees, only 12 percent of employees were in self-funded plans; in contrast, 89 percent of employees of firms that have 5,000 or more employees are in self-funded plans (Kaiser Family Foundation, Health Research and Educational Trust (HRET), and National Opinion Research Center (NORC) 2008).

We take advantage of this fact and use firm size as a proxy for self-insurance status. We create two further subsamples among the children of parents with group insurance coverage: those whose parents work for firms with less than 100 employees,

[26] We estimate that 28 percent of parents in households under 150 percent of the poverty line have group health insurance compared to 71 percent and 89 percent of parents in households between 150 and 300 percent of the poverty line and over 300 percent of the poverty line, respectively.

[27] We also find that, for those whose parents are included in a group plan, there is a decrease in nongroup coverage and an increase in group coverage. Individuals who used to pay for their own individual plans appear to drop them and join their parents' plan. This transition likely saves money for these individuals and potentially provides them with superior insurance.

[28] According to the 2008 Kaiser family Fundation, HRET, NORC *Employer Heath Benefits Survey*, 55 percent of workers who receive health insurance through their employer are in self-funded insurance plans.

Table 7—The Effect of Extended Parental Coverage Laws on the Likelihood of Living with One's Parents

	Difference-in-difference		Triple difference
	Full sample ($N = 127{,}106$)	Eligible sample ($N = 100{,}218$)	Full sample ($N = 127{,}106$)
Impact of parental coverage law	0.012*	0.019**	0.038**
	(0.007)	(0.007)	(0.010)

Notes: Each cell presents the results from separate regressions models. Control variables include state, year, and age fixed effects, the state- and year-specific unemployment rate, and individual-level covariates. Standard errors are clustered on state. The sample includes all observations between the ages of 19 and 24, using the 2001–2009 CPS.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

and those whose parents work for firms with more than 100 employees.[29] We expect to find larger effects in the smaller firms because these smaller firms are substantially more likely to be required to follow the state insurance law. In the absence of ERISA's effects, this prediction would likely be reversed, since small employers face higher expected premiums and therefore are less likely to offer health insurance (see Randall P. Ellis and Ching-to Albert Ma (forthcoming)). Indeed, we find our largest increase in private insurance coverage in the group of children whose parents have group insurance coverage through a firm with fewer than 100 employees. We also find evidence of reverse crowd-out in this subsample. The decline in public insurance coverage is considerably larger for children of parents who work in small firms, as we would expect if it were caused by the extended parental coverage laws.

At this point, we return to the issue of potential sample selection that may result from focusing the preceding analysis on just those young adults who still live with their parents. To examine this issue, we estimate a regression model analogous to those in the top panel of Table 5 except that the dependent variable is whether a young adult lives with his or her parent(s). We report these results in Table 7.

Our results suggest that extended parental coverage laws appear to increase the likelihood of living with one's parents. Consider, for instance, the difference-in-difference specification for the eligible sample. In Table 5, we estimate that extended parental coverage laws increase the likelihood of private insurance coverage by 2.2 percentage points. When we estimate the analogous model, but change the dependent variable to indicate living with one's parents, we obtain a similar effect. The introduction of an extended parental coverage law is found to statistically significantly increase the likelihood of living with one's parents by 1.9 percentage points. In fact, all of the results reported in Table 7 are similar to those for private insurance coverage for the full sample (panel A) in Table 5.

Our interpretation of these findings is that a change in the likelihood of living with one's parents is one mechanism by which these laws have an impact on health insurance coverage. Young adults may have either moved back into their parents'

[29] We also experimented with alternative firm size cut-offs (like 500 employees) and obtained qualitatively similar results.

homes or simply chosen not to move out. The fact that the point estimates in Tables 5 and 7 are similar does not necessarily mean that all individuals who were affected by the law changed their living arrangements, because there are standard errors on both estimates. It does suggest, however, that this is an important avenue through which individuals gained coverage. In fact, this mechanism makes sense because many of these laws have provisions that either directly or indirectly impose it. Some states specifically require co-residence. It is more common for states to impose the condition that the child be a dependent for tax purposes.[30] IRS guidelines indicate that the child must live with the parent for more than half of the tax year to satisfy this requirement.

This finding does, however, have implications for our interpretation of the findings in Table 6, where we impose the sample restriction that young adults must live with one's parents. It suggests that it would be inappropriate to interpret the results in Table 6 as the magnitude of the true policy effects. However, it is appropriate to use these results as specification checks.[31]

This discussion also highlights another important implication regarding the estimated magnitude of ability of extended parental coverage laws to reduce rates of uninsurance. Earlier, we interpreted our results in Table 5 as suggesting that if all 51 states (including DC) enacted an extended parental coverage law, the rate of uninsurance would decline 11 percent for young adults. This 11 percent estimate is based on policies that affect only the 45 percent whose private insurance plans are not self-funded. A national law would not have this limitation and would, therefore, reduce the rate of uninsurance by 24 percent (11/0.45) for those eligible.

V. Discussion

The results of our analysis suggest that both the introduction of SCHIP and extended parental coverage laws were effective in increasing rates of insurance coverage for older teens and young adults, at least for certain population subgroups. SCHIP was particularly effective at increasing insurance coverage for older teens with incomes below 150 percent of the poverty line without evidence of substantial crowd-out. Extended parental coverage laws had a sizeable impact on the insurance coverage of nonstudents, young adults between 150 and 300 percent of the poverty line, and those whose parents worked for relatively small firms that had group insurance plans. We also found suggestive evidence that the increase in insurance coverage that this policy generated may have been accompanied by a reduction in public insurance coverage (reverse crowd-out).

[30] Getting an exact count of which states are in each category is difficult because of the complexity of some of the laws. For instance, in Massachusetts, IRS dependents can be included on their parents' plan. If they are no longer IRS dependents, but have been within the past two years, they can still be covered, but a tax liability is imposed on the value of the benefit.

[31] In an additional specification check, we conducted a falsification exercise where we replicated our analysis for a sample of 16–18-year-olds, who always should be eligible for parental coverage. In this exercise, the coefficients were uniformly smaller and generally statistically insignificant at conventional levels, although these estimates were less precise. We view this falsification exercise as providing some additional support for the validity of our empirical strategy, although its power is somewhat limited.

Determining whether policymakers should endorse extending either or both of these programs requires a weighing of their advantages and disadvantages as implemented in the past and as they would be extended going forward. In the case of SCHIP, for instance, our policy experiment addressed the discontinuity of coverage at age 19 created when SCHIP was introduced. An expansion of this policy could increase the maximum age of coverage to, say, 22 or 25 years of age. It is not clear whether the impact of such an extension would be the same as that of the earlier policy.

One issue to consider would be differences in crowd-out. If anything, our prior is that crowd-out would be smaller for this group than it is for younger children and, in fact, our estimates of crowd-out based on the introduction of SCHIP are consistent with the smaller of the crowd-out estimates from earlier Medicaid expansions. A potential explanation for this is that the "marginal child" gaining coverage in SCHIP is a teen who is lower in the income distribution than the younger children who gained coverage in earlier Medicaid expansions.[32] Uninsured young adults would similarly be more likely to be drawn from the lower parts of the income distribution and, if our hypothesis is correct, crowd-out may be less important for this group.

Another factor that may suggest even larger effects for young adults is that they have fewer options for coverage. Young children have the ability to be included in their parents' plan if their parents are covered; parental coverage is quite high for younger children as shown in Figure 2. Young adults would have to rely on their own private coverage as their alternative to a public plan, yet private coverage rates are relatively low for young adults (also shown in Figure 2).

To provide a ballpark estimate of the number of additional young adults who may be covered if SCHIP were extended to cover individuals up to their twenty-second or twenty-fifth birthday, we ignore these caveats for the moment. We assume that the impact on the incremental groups covered is the same as we have estimated for those at age 19 with the same extent of crowd-out. Based on our results in Table 3, we apply the aggregate estimate that insurance coverage of any type would rise by 3.2 percentage points to simulate this statistic. We combine this value with population estimates from the 2005–2007 American Community Surveys indicating that there are about 13.3 million individuals between the ages of 19 and 21 and another 12.2 million between the ages of 22 and 24. If 3.2 percentage points more members of these groups received health insurance in response to an SCHIP expansion, this would increase coverage by 426,000 and 390,000 individuals, respectively.

We also hypothesize that extended parental coverage laws would have larger effects than those we have estimated if they were implemented at the federal level. A key constraint on the effectiveness of these laws as currently enacted at the state level is that ERISA significantly restricts the target population. As discussed earlier, 55 percent of those parents who have employer-provided health insurance are covered by firms that self-insure, and ERISA excludes those plans from state regulation. If implemented at a national level, that restriction would not exist. Moreover, state

[32] We draw this conclusion based on our calculations of the average characteristics of children who are eligible under a new set of policies, but not eligible under the previously existing policies. For the Medicaid expansions of the late 1980s and early 1990s, we find that newly eligible children averaged 5 years old and they were at 80 percent of the poverty line. For SCHIP, the comparable statistics were 11 years old and 140 percent of the poverty line.

level plans require children to reside in the same state as the parents and that would not need to be true if the policy were implemented at the federal level.

To simulate the effect of a national extended parental coverage law for those aged 19 to 24, we apply our estimates indicating that uninsurance would decline by 3.3 percentage points or 7.3 percentage points, where the distinction is based on whether individuals covered by self-funded private plans are exempt or not. With 25.5 million individuals between the ages of 19 and 24, our estimates of the impact of the policy imply that 842,000 or 1.9 million would fall from the ranks of the uninsured, depending on whether the laws were implemented at the state or national level. This would include those who have already received coverage as a result of the state laws.

The question of who pays for the insurance coverage is also a relevant issue. We find some support for the notion that extended parental coverage laws generate reverse crowd-out in that some individuals may drop public insurance coverage for what may be superior private insurance coverage. This shifts the cost of the coverage from the taxpayer to the individual's family. If the employer is unable to perfectly shift the increased premium costs onto the individual's family, the costs may be borne by the employer or by other employees at the firm. To the extent that the family pays, such an outcome may be optimal, but to the extent that the costs are shifted to others, it may not be.

In the end, policymakers will be required to weigh these trade-offs in determining whether to introduce either or both of these policies. Nevertheless, our analysis has provided evidence indicating that both types of policies as implemented have increased insurance coverage among the targeted groups.

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